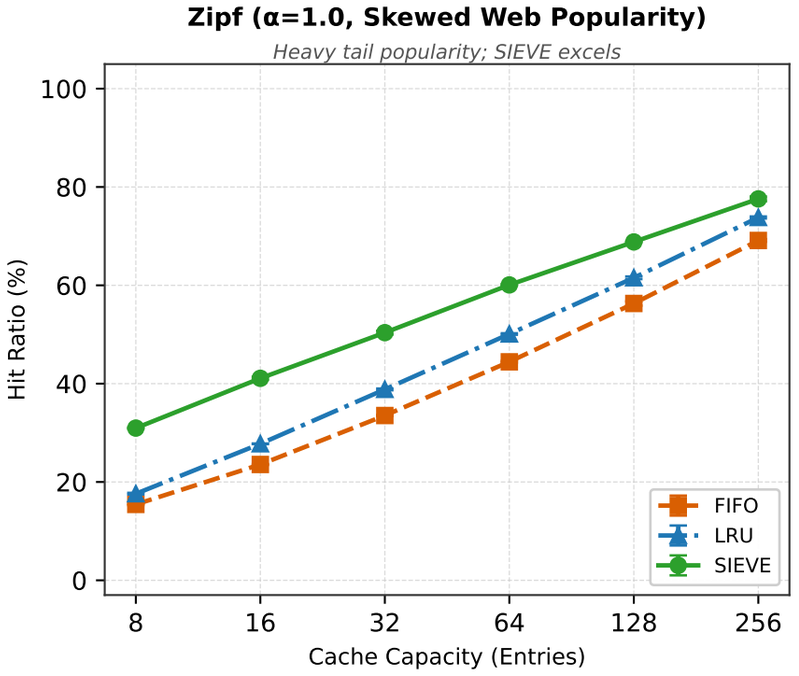
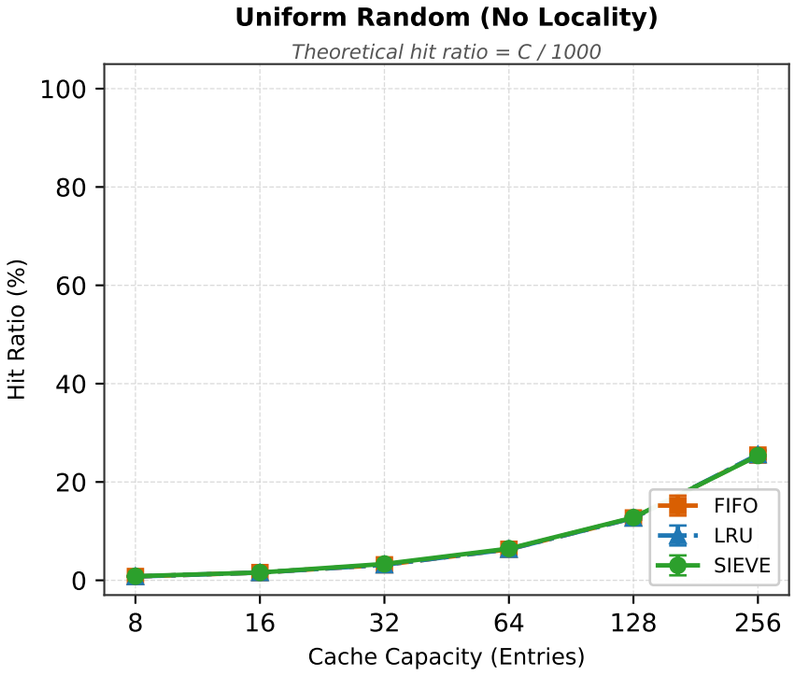
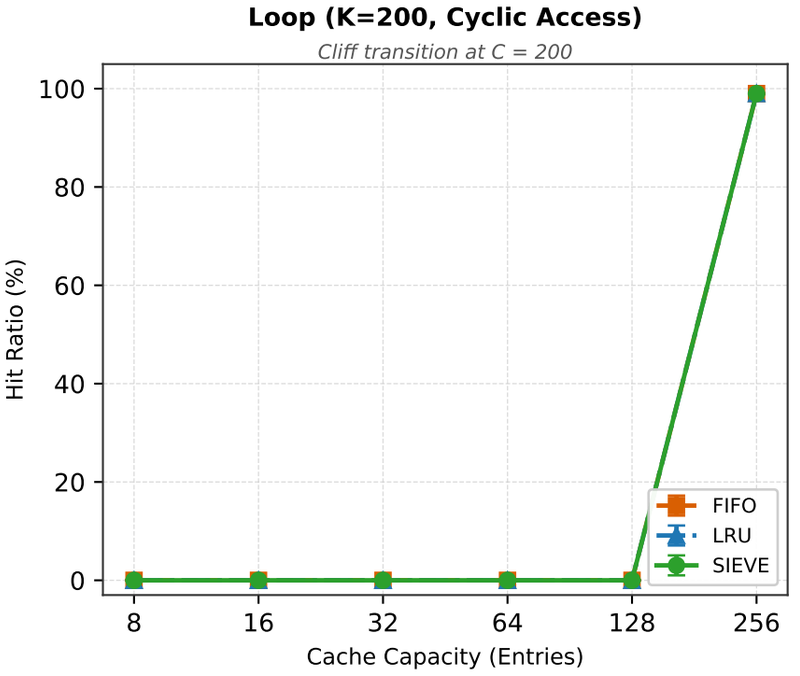
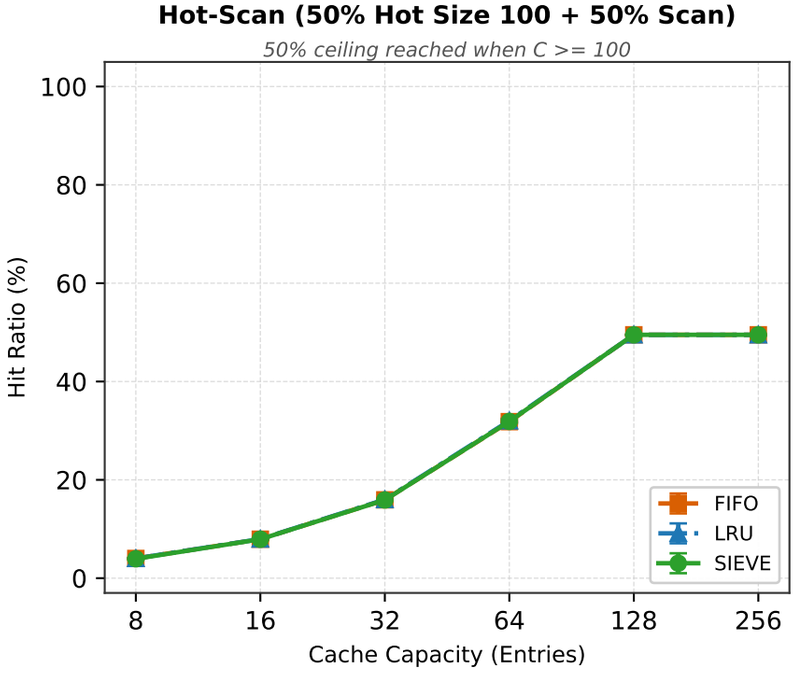
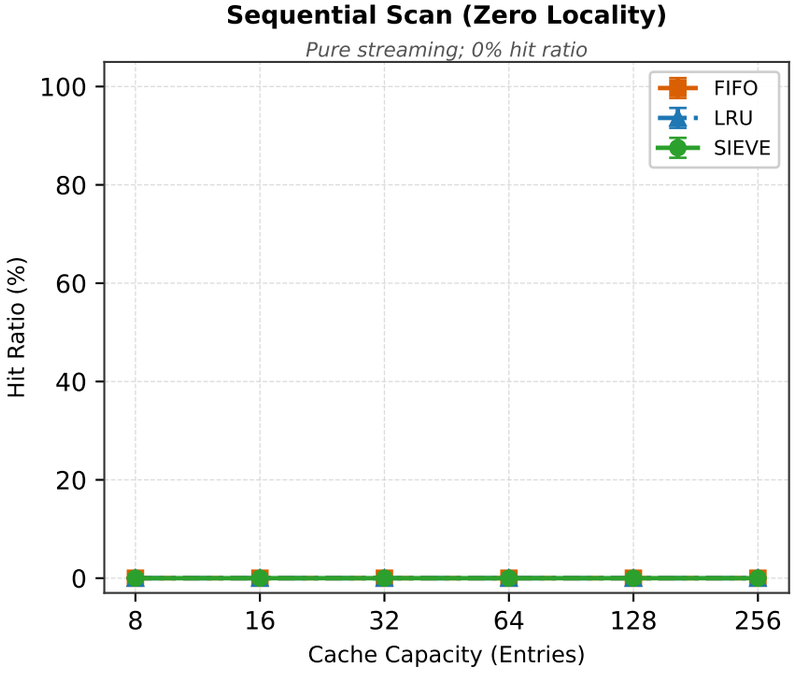
Cache Hit Ratio vs Capacity Across Synthetic Workloads












```
Evaluation Matrix Summary
─────────────────────────────────────

• Simulator: Single-threaded C11 user-space
• Replay: Exact same trace per capacity
• Capacities: 8, 16, 32, 64, 128, 256
• Seeds: 42, 1337, 99991 (stochastic)
• Error bars: ±1 standard deviation

Core Algorithmic Takeaways:
─────────────────────────────────────

1. Zipf Skew: SIEVE achieves +3.8% to +13.4%
   higher hit ratio than LRU across all C.
2. Working Set Cliff: Loop transitions abruptly
   from 0% (C < 200) to 99% (C = 256).
3. Scan Pollution: All policies suffer 0% hits
   under pure sequential scanning.
4. Uniform: All policies track theoretical C/K.
```